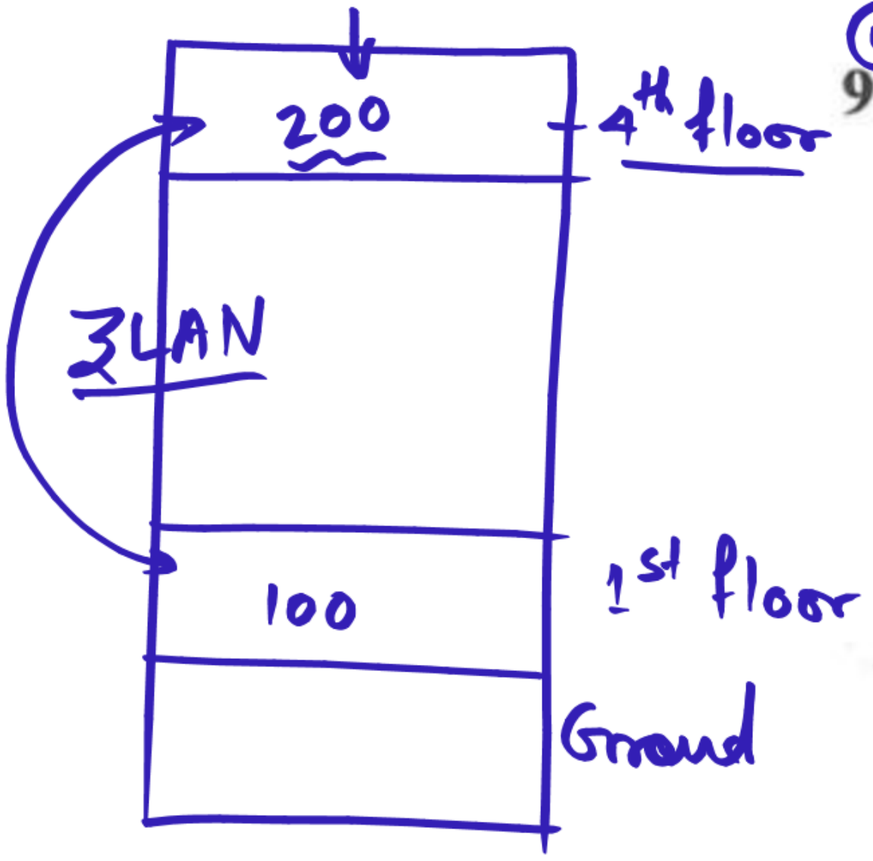
Random Test

Computer Network

Rajasthan - RSSB & RPSC



91 In a Data Entry Office, the computer centers are on 1st and 4th floor. Both the centres are connected three LAN. However the signals take time to travel. Which amongst the following hardware can improve the data transmission?

(A) Modem (B) Repeater
(C) Router (D) Computer
(E) Question not attempted

एक डेटा एंट्री कार्यालय में, कंप्यूटर केन्द्र पहली और चौथी मंजिल पर हैं। दोनों केन्द्र LAN के माध्यम से जुड़े हुए हैं। सिग्नलों (संकेतों) को पहुँचने में समय लगता है। निम्नलिखित में से कौन सा हार्डवेयर डाटा संचरण में सुधार कर सकता है?

(A) मॉडेम (B) रिपीटर
(C) राउटर (D) कंप्यूटर
(E) अनुत्तरित प्रश्न

88 All the information on the internet is passed using _____ switching method.

(A) Portable (B) Packets

(C) Pouch (D) Primary

(E) Question not attempted

इंटरनेट पर सभी जानकारियाँ _____ विधि द्वारा होकर गुजरती है।

(A) पोर्टेबल (सुवाह्यता) (B) पैकेट

(C) पाउच (D) प्राथमिक

(E) अनुत्तरित प्रश्न

104 In LAN, connectivity can be created using Wi-Fi, or _____ cables.

(A) TV (B) Radio

(C) Mobile (D) Ethernet

(E) Question not attempted

LAN में Wi-Fi, या _____ केबल की मदद से सम्पर्क स्थापित किया जा सकता है।

(A) TV (B) रेडियो

(C) मोबाइल (D) इथरनेट

(E) अनुत्तरित प्रश्न

108 _____ signals are discrete and generated by digital modulations.

(A) Analog (B) Digital

(C) Noise (D) Bandwidth

(E) Question not attempted

_____ सिग्नल असंतत है तथा उसे डिजिटल प्रतिरूपण से प्राप्त किया जाता है।

(A) सदृश (B) डिजिटल

(C) शोर (D) बैंडविथ

(E) अनुत्तरित प्रश्न

110 _____ is a network security system designed to protect a trusted private network from an untrusted outside network to which it is connected.

(A) Bridge (B) Cookies

(C) Firewall (D) SSL

(E) Question not attempted

_____ एक नेटवर्क सुरक्षा प्रणाली है जिसे विश्वस्त निजी नेटवर्क को, अविश्वासी बाहरी नेटवर्क जिससे यह जुड़ा है, की सुरक्षा के लिए बनाया गया है।

(A) ब्रिज (B) कुकीज़

(C) फायरवॉल (D) SSL

(E) अनुत्तरित प्रश्न

IP Security

↳ IP Encapsulation

&

Tunneling

Virtual Private Network

117 IPSEC, IP Encapsulation and Tunneling are the technologies used in _____ only.

(A) Intranet (B) TCP

(C) VPN (D) Internet

(E) Question not attempted

IPSEC, IP संक्षिप्तीकरण तथा टनलिंग ऐसी तकनीक हैं, जिनका प्रयोग _____ में होता है।

(A) इंट्रानेट (B) TCP

(C) VPN (D) इंटरनेट

(E) अनुत्तरित प्रश्न

23. Match TCP/IP layers and protocols -

Column – I

3 (P) Transport layer → UDP

1 (Q) Network (internet) layer → ICMP

2 (R) Application layer → FTP

Column – II

1. ICMP
2. FTP
3. UDP

(A) P-2, Q -1, R-3 **(B)** P-1, Q-2, R-3

(C) P-3, Q-1, R-2 **(D)** P-3, Q-2, R-1

19. Which protocol layer uses the protocol SMTP, HTTP, FTP?

(A) Application layer (B) Transport layer

(C) Internal layer (D) Hardware layer

SMTP, HTTP, FTP प्रोटोकॉल का इस्तेमाल कौनसा प्रोटोकॉल लेयर करता है?

(A) एप्लीकेशन लेयर (B) ट्रान्सपोर्ट लेयर

(C) इंटरनल लेयर (D) हार्डवेयर लेयर

Quantizing.

Pulse Code Modulation

39. Quantizing noise occurs in -

(A) PCM (B) TDM

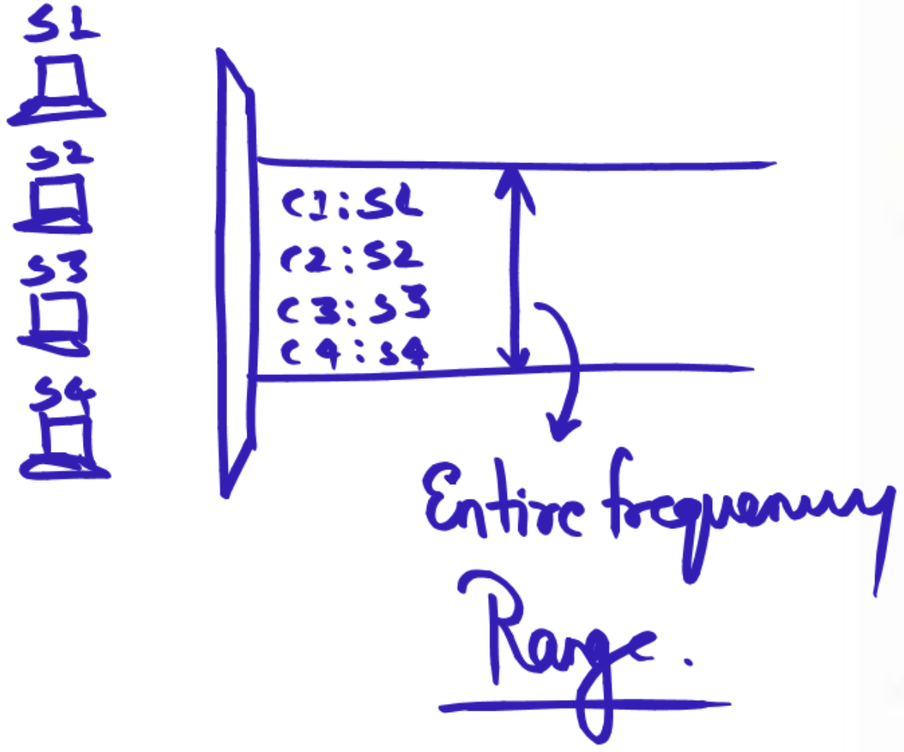
(C) FDM (D) PPM

परिमाणीकरण शोर होता है –

(A) PCM में (B) TDM में

(C) FDM में (D) PPM में

Code Division Multiple Access



CDMA More Secure Compare to GSM.

83. Select the false statement about CDMA technology –

I. CDMA technology optimizes the use of available bandwidth as it transmits over the entire frequency range.

II. Each user in a CDMA system uses a different code to modulate their signal.

III. CDMA technology offers less security as compared to the GSM technology.

IV. A SIM card is always required for the working of a CDMA device.

(A) Only (I)
(B) Only (II)
(C) (I) and (II)
(D) (III) and (IV)

93. Network devices ..... and ..... are used at physical layer.

(A) Gateway, Bridge

(B) Router, Repeater

(C) Hub, Switch

(D) Repeater, Hub

64

46. An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be -

(1) 255.255.0.0

(2) 255.255.252.0

(3) 255.255.64.0

(4) 255.255.128.0

(5) Question not attempted

class B ⇒ 16 bit Net_id | 16 bits Host bit

$64 \Rightarrow 2^6 = 64$

11111100.00000000

252.0

47. The physical layer is responsible for -

(1) Process to process delivery

(2) Bit-by-bit delivery

(3) Application to application delivery

(4) Port to port delivery

(5) Question not attempted

49. Which of the following Multiple Access Protocols is not suitable for wired networks?

(1) CSMA/CA

(2) CSMA/CD

(3) TGMA

(4) TDMA

(5) Question not attempted

CSMA/CA → Carrier Sense Multiple Access
↳ Collision Avoidance

Wireless

52. For Stop-and-Wait ARQ, for 10 data packets sent, ...acknowledgements are needed.

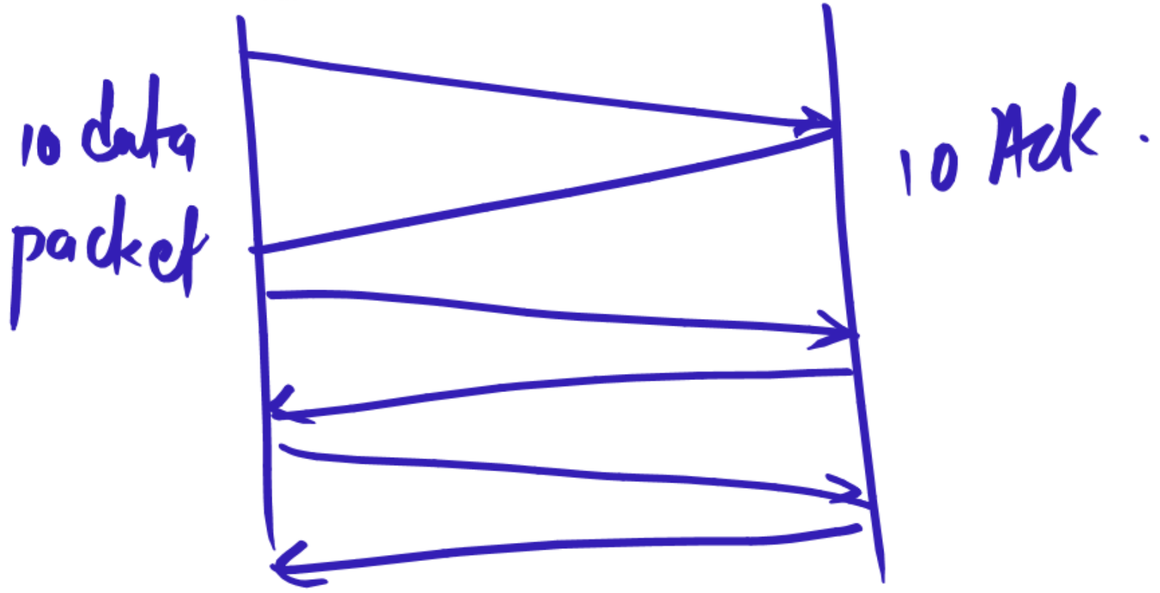
(1) 11

(2) 9

(3) 10

(4) 20

(5) Question not attempted



→ Domain Name System.

54. What is the role of DNS? It is best described as -

(1) translate a host's IPv6 address to an IPv4 address

(2) maps a MAC address to an IP address

(3) translate a host's private IP address to a public IP address

(4) maps a domain name to an IP address

(5) Question not attempted

56. If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(1) 1023
(2) 1022
(3) 1024
(4) 2046
(5) Question not attempted

Class B → 255.255.248.0

16 bit Net-id

11111000.00000000

11

Host = $2^{11} - 2$ ⇒

$2048 - 2 = 2046$

57. The Address Resolution Protocol (ARP) is used for -

(1) finding the IP address from the DNS.

(2) finding the IP address that corresponds to a MAC address.

(3) finding the IP address of the default gateway.

(4) finding the MAC address that corresponds to an IP address.

(5) Question not attempted

61. What network property is achieved through the use of the Automatic Repeat reQuest (ARQ) mechanism?

(1) Reliability

(2) Availability

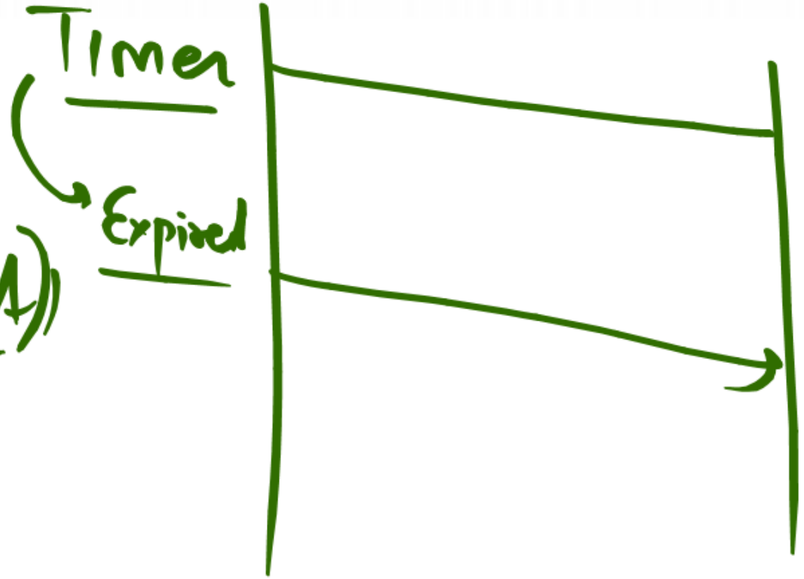
(3) Confidentiality

(4) Scalability

(5) Question not attempted

Flow Control

① Stop & wait ARQ (Automatic Repeate Request)



64. The size of the 'total length field' of IPv4 datagram is -

(1) 16 bits

(2) 4 bits

(3) 8 bits

(4) 32 bits

(5) Question not attempted

65. Which of the following is not related to data link layer?

(1) Congestion control

(2) Error control

(3) Framing

(4) MAC addressing

(5) Question not attempted

Internet Control Message Protocol.

71. ICMP protocol is a ..... layer protocol.

(1) Data link

(2) Application

(3) Transport

(4) Network

(5) Question not attempted

classless Inter Domain Routing

72. What does /24 denote in CIDR?

/24 CIDR

(1) 24 bits are used for network portion

(2) 24 addresses are available in the subnet

(3) 24 subnets are created

(4) 24 bits are used for host portion

(5) Question not attempted

24

$2^{32-24} = 2^8 = 256$

HostIP = 256 - 2 = 254

73. Network Allocation Vector (NAV) is associated with which of the following MAC technologies?

(1) CSMA/CD

(2) Bluetooth

(3) Wi-Fi

(4) Ethernet

(5) Question not attempted

CSMA/CA
II
Wireless

80. Which of the following is true for Circuit Switching handling multiple simultaneous connections?

(1) A single path is used and shared for all data

(2) It encodes all data into a single channel

(3) Separate physical paths are allocated for each channel

(4) All connections are multiplexed into a single channel

(5) Question not attempted

82. Which of the following protocol may be used as zeroconf and plug-and-play?

(1) DHCP

(2) ICMP

(3) NAT



(4) TCP

(5) Question not attempted

zeroconf ⇒ DHCP client / server

↓

0.0.0.0.

84. The reliable data transfer is implemented by.....

(1) Network layer

(2) Presentation layer

(3) Physical layer

(4) Transport layer

(5) Question not attempted

TCP

94. How many bits for 'Fragment Offset' field are there in IPv4 header?

(1) 10 bits

(2) 32 bits

(3) 13 bits

(4) 16 bits

(5) Question not attempted

95. The size of MAC address (in bits) is -

(1) 24

(2) 48

(3) 36

(4) 42

(5) Question not attempted

98. The message 1001 is to be transmitted using the CRC polynomial $x^3 + x + 1$ to protect it from errors. The message that should be transmitted is -

(1) 1001000 **(2)** 1001100

(3) 1001110 **(4)** 1001011

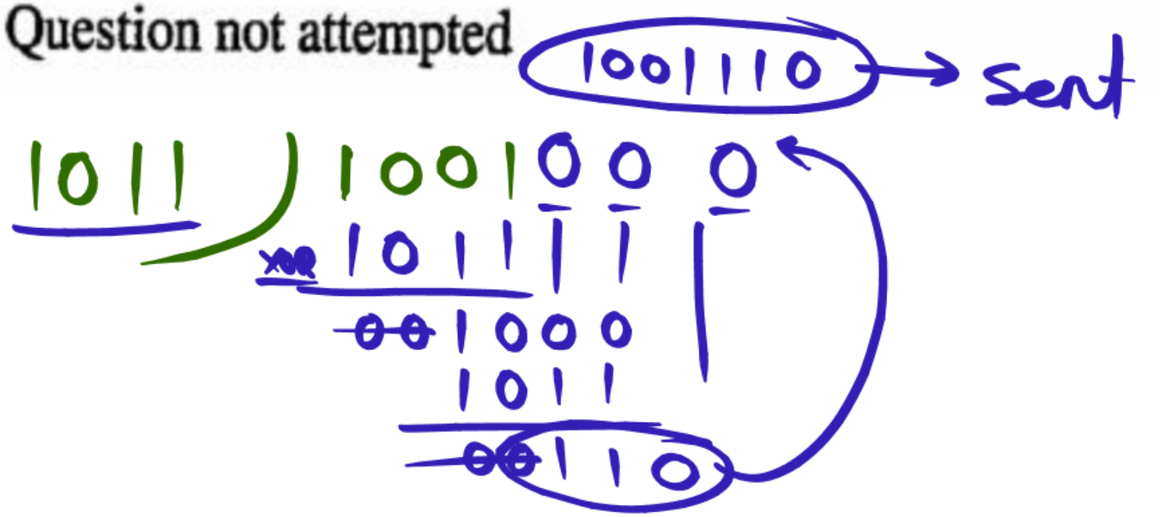
(5) Question not attempted

CRC generator

$x^3 + x + 1$

$1.x^3 + 0.x^2 + 1.x^1 + 1.x^0$

1011



99. The three-way handshake mechanism in TCP aims to -

(1) secure the communication.

(2) send bits from one device to another over an optical fibre connection.

(3) perform error detection and correction.

(4) establish connection and synchronize sequence numbers.

(5) Question not attempted

45. If the channel band is limited to 6 kHz and signal to noise ratio is 16, what would be the capacity of channel?

(A) 15.15 kbps (B) 24.74 kbps

(C) 30.12 kbps (D) 52.18 kbps

यदि चैनल 6 kHz तक सीमित है और सिग्नल से शोर अनुपात 16 है, तो चैनल की क्षमता क्या होगी?

(A) 15.15 kbps (B) 24.74 kbps

(C) 30.12 kbps (D) 52.18 kbps

channel

Noiseless channel

Noisy channel

Signal to Noise Ratio

$B \log_2(1+SNR)$

24000

$6000 \log_2(1+16)$

$6000 \log_2(17)$

4.1..

② $\log_2(4)$

③ $\log_2(8)$

④ $\log_2(16)$